

Design Tokens: Plugin Slots but for your CSS

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Slides and materials available

<https://github.com/brian-smith-tcrl/design-tokens-talk-2026>



What are design tokens?

What are design tokens?

“Design tokens are indivisible pieces of a design system such as colors, spacing, typography scale.”

- Design Tokens W3C Community Group (DTCG)
<https://www.designtokens.org/>

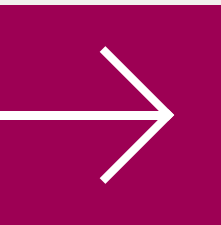
What are design tokens?

"Design tokens are indivisible pieces of a design system such as colors, spacing, typography scale."

- Design Tokens W3C Community Group (DTCG)

<https://www.designtokens.org/>





Before plugin slots

Before plugin slots: MFEs as all-in-ones



Before plugin slots: MFEs as all-in-ones

All-in-ones do everything you need...



Before plugin slots: MFEs as all-in-ones

All-in-ones do everything you need...

...until you want to customize something...



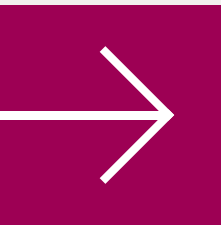
Before plugin slots: MFEs as all-in-ones

All-in-ones do everything you need...

...until you want to customize something...

...and you have to replace it (fork).





Plugin slots

Plugin slots: Extensible MFEs



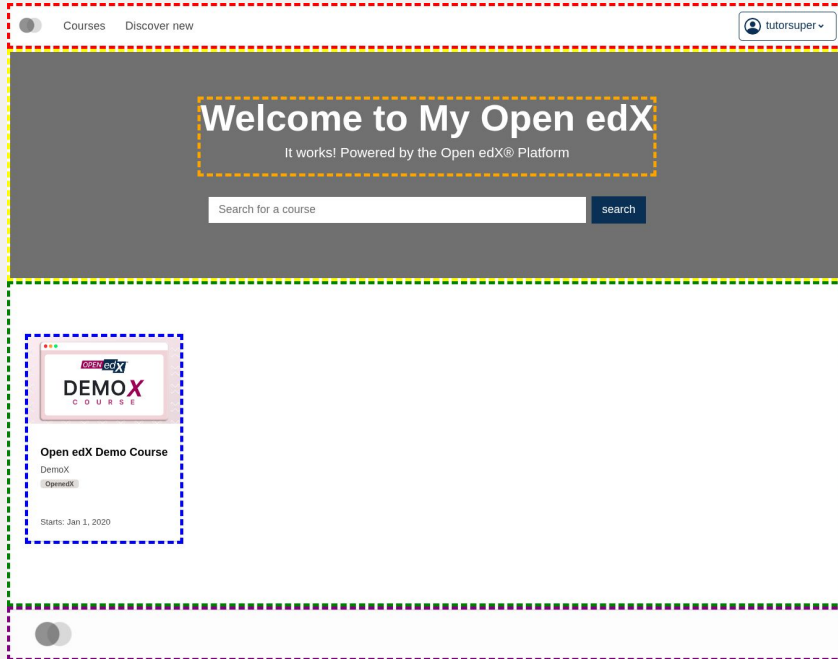
Plugin slots: Extensible MFEs

With plugin slots...



Plugin slots: Extensible MFEs

With plugin slots...



<https://github.com/brian-smith-tcril/design-tokens-talk-2026/tree/main/plugin-catalog-slot-highlighter>



Plugin slots: Extensible MFEs

With plugin slots...

...you can replace components!

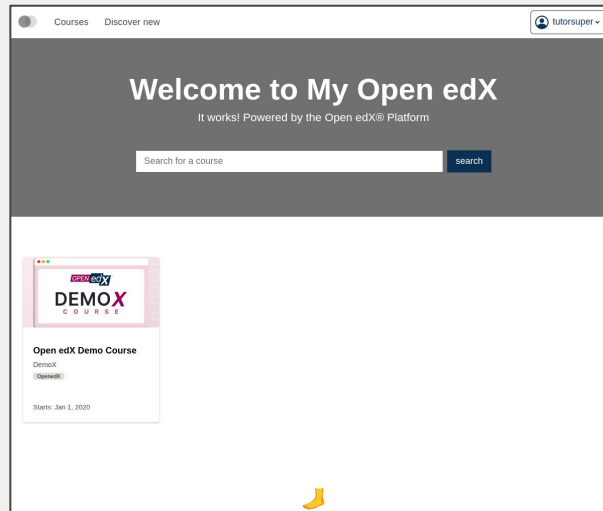


Plugin slots: Extensible MFEs

With plugin slots...

...you can replace components!

```
const config = {  
  pluginSlots: {  
    'org.openedx.frontend.layout.footer.v1': {  
      plugins: [  
        {  
          // Insert a custom footer  
          op: PLUGIN_OPERATIONS.Insert,  
          widget: {  
            id: 'custom_footer',  
            type: DIRECT_PLUGIN,  
            RenderWidget: () => (  
              <h1 style={{textAlign: 'center'}}>👉</h1>  
            ),  
          },  
        ],  
      },  
    },  
  },  
}
```

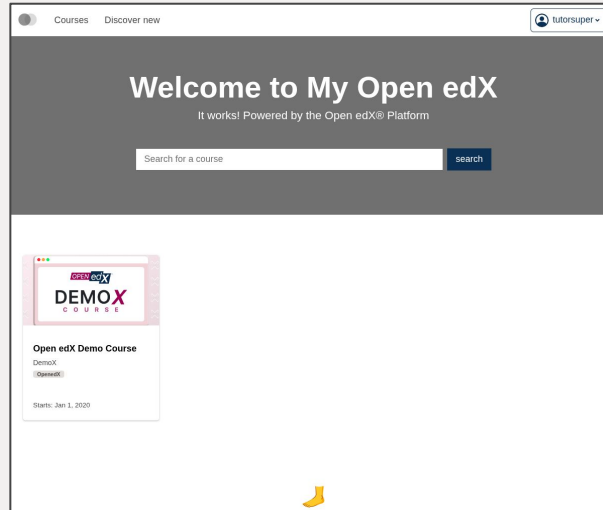


Plugin slots: Extensible MFEs

With plugin slots...

...you can replace components*

```
const config = {
  pluginSlots: {
    'org.openedx.frontend.layout.footer.v1': {
      plugins: [
        {
          // Insert a custom footer
          op: PLUGIN_OPERATIONS.Insert,
          widget: {
            id: 'custom_footer',
            type: DIRECT_PLUGIN,
            RenderWidget: () => (
              <h1 style={{textAlign: 'center'}}>👉</h1>
            ),
          },
        },
      ],
    },
  },
}
```



*at build time



Plugin slots: Extensible MFEs

Wait!



Plugin slots: Extensible MFEs

Wait!

Why are we talking about plugin slots?



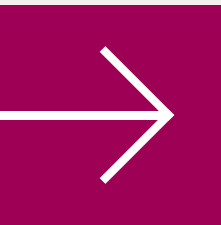
Plugin slots: Extensible MFEs

Wait!

Why are we talking about plugin slots?

I thought this was a design tokens talk!





Design tokens

Design tokens



Design tokens

How do tokens fit into this?



Design tokens

How do tokens fit into this?

Design tokens are knobs!



Design tokens

What knobs exist in the Open edX® platform?



Design tokens

What knobs exist in the Open edX[®] platform?

Paragon has a bunch!

[https://github.com/openedx/paragon/\[...\]/tokens/src](https://github.com/openedx/paragon/[...]/tokens/src)

`--pgn-color-brand-base`



Design tokens

What knobs exist in the Open edX[®] platform?

Paragon has a bunch!

[https://github.com/openedx/paragon/\[...\]/tokens/src](https://github.com/openedx/paragon/[...]/tokens/src)

```
$ npm run build-tokens -- --verbose
```

```
[...]
```

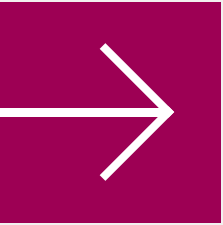
```
core: processed 660 tokens
```

```
[...]
```

```
theme (light): processed 2712 tokens (660 core, 2052 theme)
```

<https://github.com/openedx/paragon/commit/36268ca>





Paragon tokens

Paragon tokens



Paragon tokens

Paragon design tokens...

--pgn-color-brand-base



Paragon tokens

Paragon design tokens...

```
{  
  "color": {  
    "$type": "color",  
    [...]  
    "brand": {  
      [...]  
      "base": {  
        "source": "$brand",  
        "actions": {  
          "default": "{color.action.default.brand.base}"  
        },  
        "$value": "#9D0054",  
        "$description": "Basic brand color."  
      },  
    },  
  },  
  [...]  
}
```

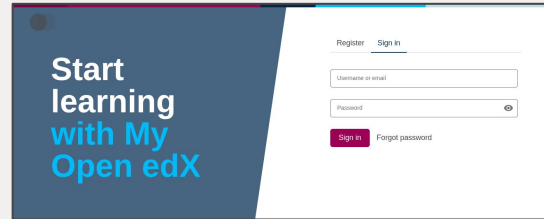
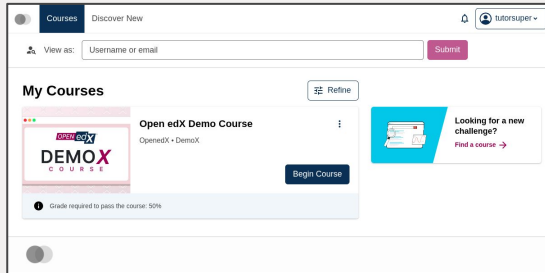
[https://github.com/openedx/paragon/blob/\[...\]/tokens/src/themes/light/global/color.json#L640-L648](https://github.com/openedx/paragon/blob/[...]/tokens/src/themes/light/global/color.json#L640-L648)

--pgn-color-brand-base



Paragon tokens

Paragon design tokens style the entire platform...



--pgn-color-brand-base



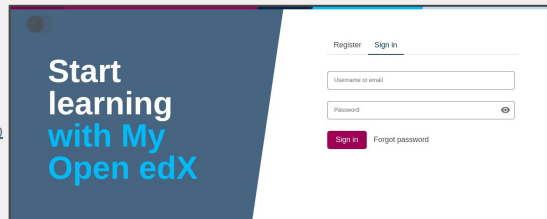
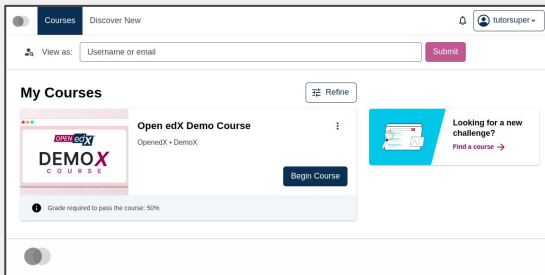
Paragon tokens

Paragon design tokens style the entire platform...

```
.extra-large-screen-top-stripe {
  display: flex;
  height: 0.5rem;
  background-image: linear-gradient(
    102.02deg,
    var(--pgn-color-brand-700) 10%,
    var(--pgn-color-brand-base) 10%,
    var(--pgn-color-brand-base) 45%,

```

https://github.com/openedx/frontend-app-authn/blob/1.../src/sass/_base_component.scss#L135-L150



```
"color": {
  "$type": "color",
  "btn": {
    ...
  }
  "bg": {
    "brand": {
      "source": "$btn-brand-bg",
      "$value": "{color.brand.base}"
    }
  }
}
```

<https://github.com/openedx/paragon/blob/1.../tokens/src/themes/light/components/Button/brand.ison#L25-L28>

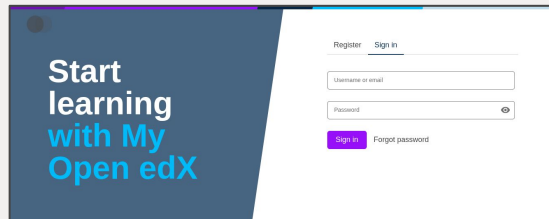
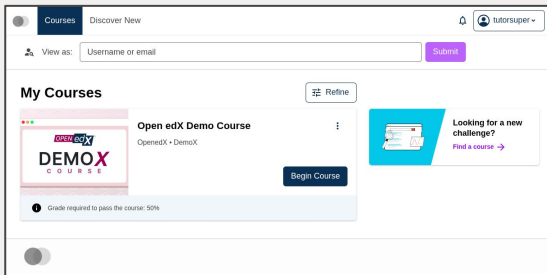
`--pgn-color-brand-base`



Paragon tokens

Paragon design tokens style the entire platform...

...with your brand!



```
{  
  "type": "color",  
  "color": {  
    "brand": {  
      "base": {  
        "value": "#9810fa",  
        "source": "$brand",  
        "description": "Basic brand color."  
      }  
    }  
  }  
}
```

<https://github.com/openedx/sample-plugin/tree/1.1/brand-sample>

--pgn-color-brand-base

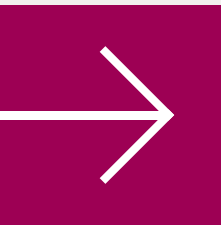


Paragon tokens

Paragon design tokens style the entire platform...

...but what about knobs that aren't in Paragon?





Application tokens

Application tokens



Application tokens

Some tokens only make sense for one app

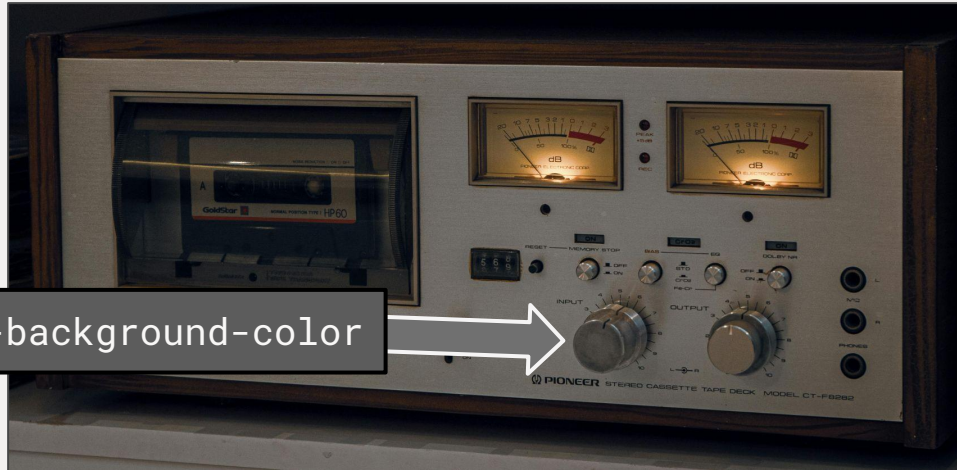


Application tokens

Some tokens only make sense for one app

...so we can add app-specific knobs

--catalog-home-page-banner-background-color

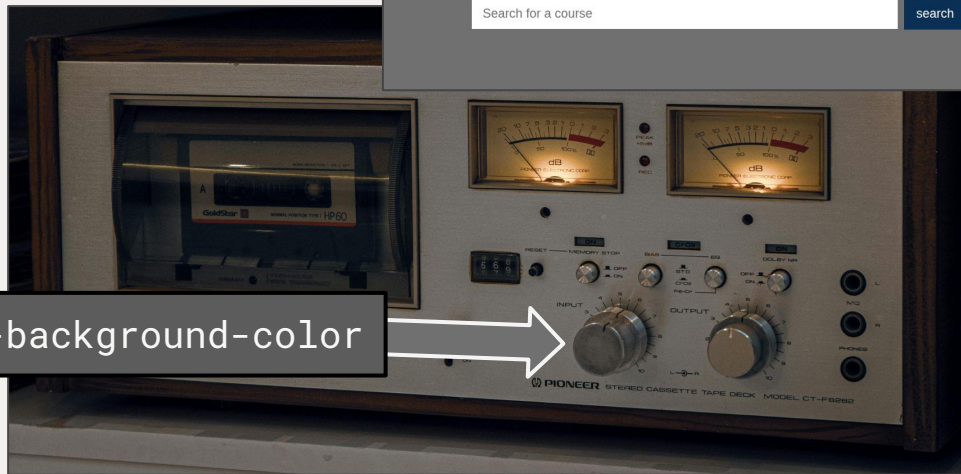
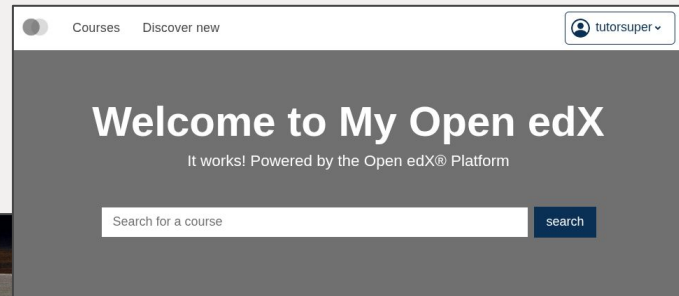


Application tokens

Some tokens only make sense for one app

...so we can add app-specific knobs

...directly to MFEs



`--catalog-home-page-banner-background-color`

Application tokens

Some tokens only make sense for one app

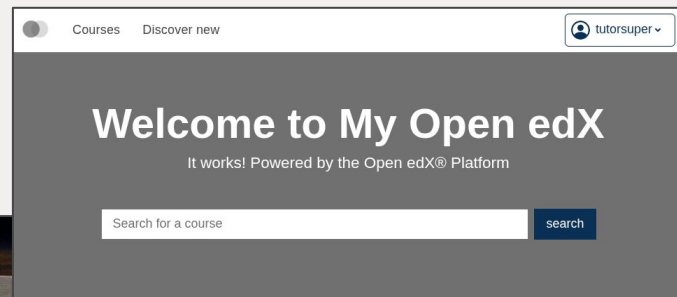
...so we can add app-specific knobs

...directly to MFES

```
.home-banner {  
  &::before {  
    [...]  
    background-color: var(--catalog-home-page-banner-background-color, var(--pgn-color-gray-500));  
  }  
}
```

[https://github.com/openedx/frontend-app-catalog/blob/\[...\]/src/home/components/home-banner/index.scss#L14](https://github.com/openedx/frontend-app-catalog/blob/[...]/src/home/components/home-banner/index.scss#L14)

`--catalog-home-page-banner-background-color`



Application tokens

Some tokens only make sense for one app

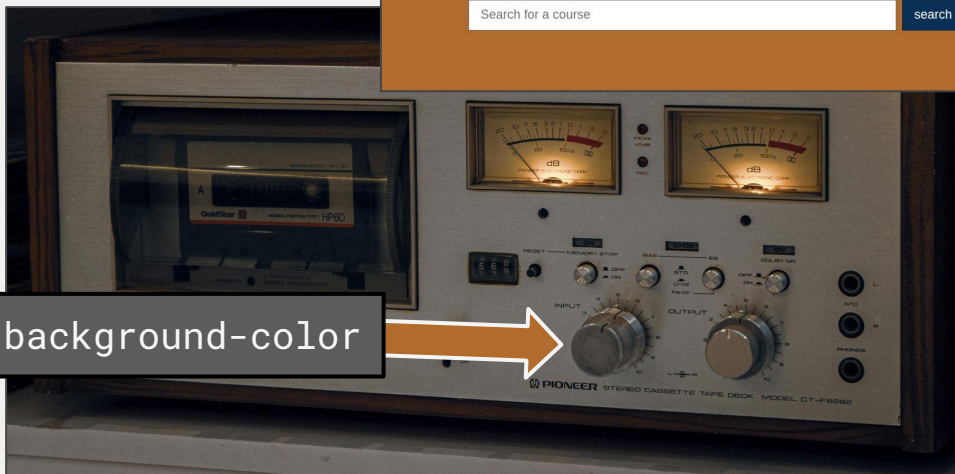
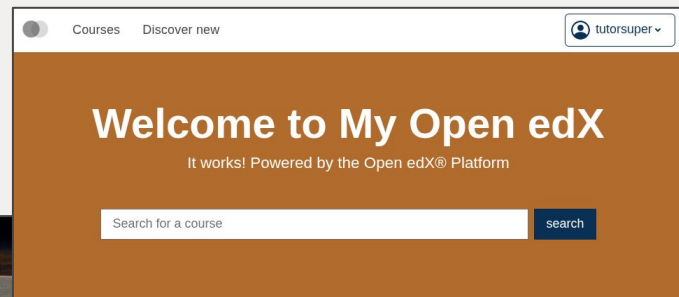
...so we can add app-specific knobs

...directly to MFEs

```
{  
  "catalog": {  
    "home-page": {  
      "banner": {  
        "background-color": {  
          "$value": "#b16b2c",  
          "$type": "color"  
        }  
      }  
    }  
  }  
}
```

<https://github.com/brian-smith-tcrl/design-tokens-talk-2026/tree/main/brand-catalog-background>

--catalog-home-page-banner-background-color



Application tokens

Some tokens only make sense for one app

...so we can add app-specific knobs

...directly to MFEs

...or...



Application tokens

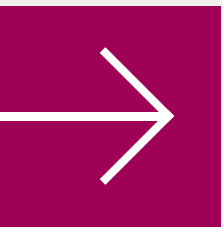
Some tokens only make sense for one app

...so we can add app-specific knobs

...directly to MFEs

...or to plugins!

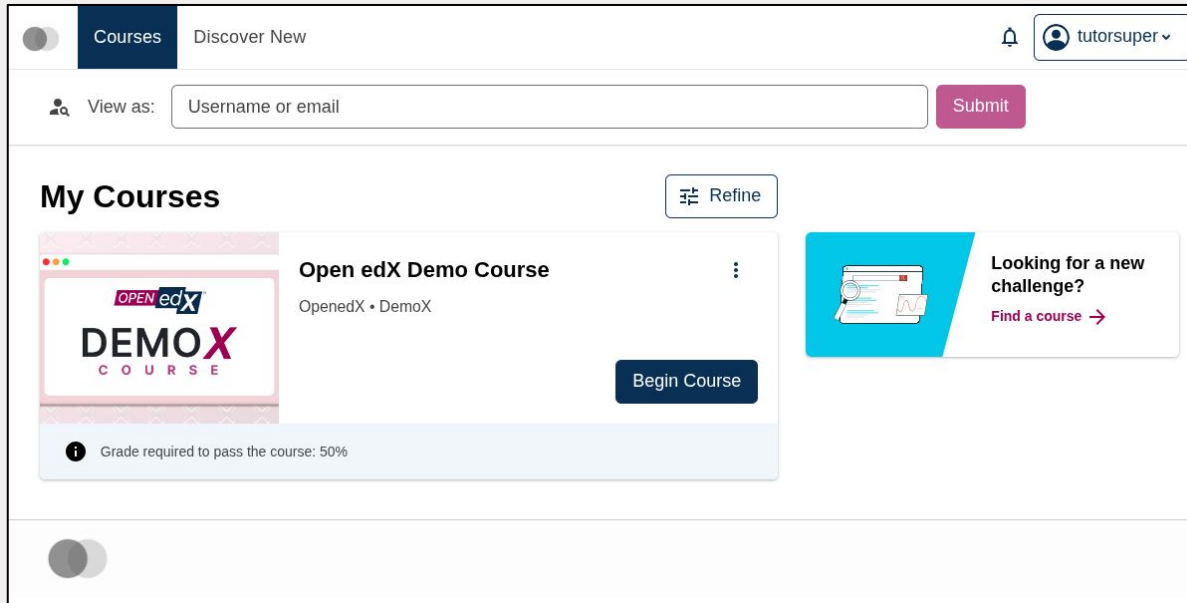




Application tokens in plugins

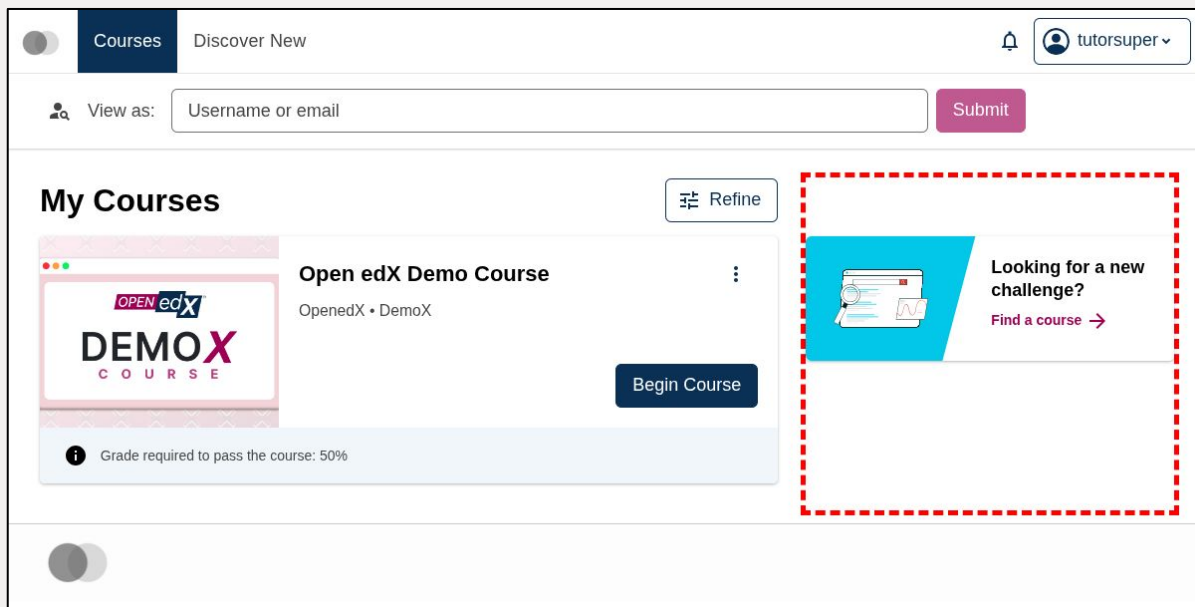
Application tokens in plugins

Application tokens in plugins



Application tokens in plugins

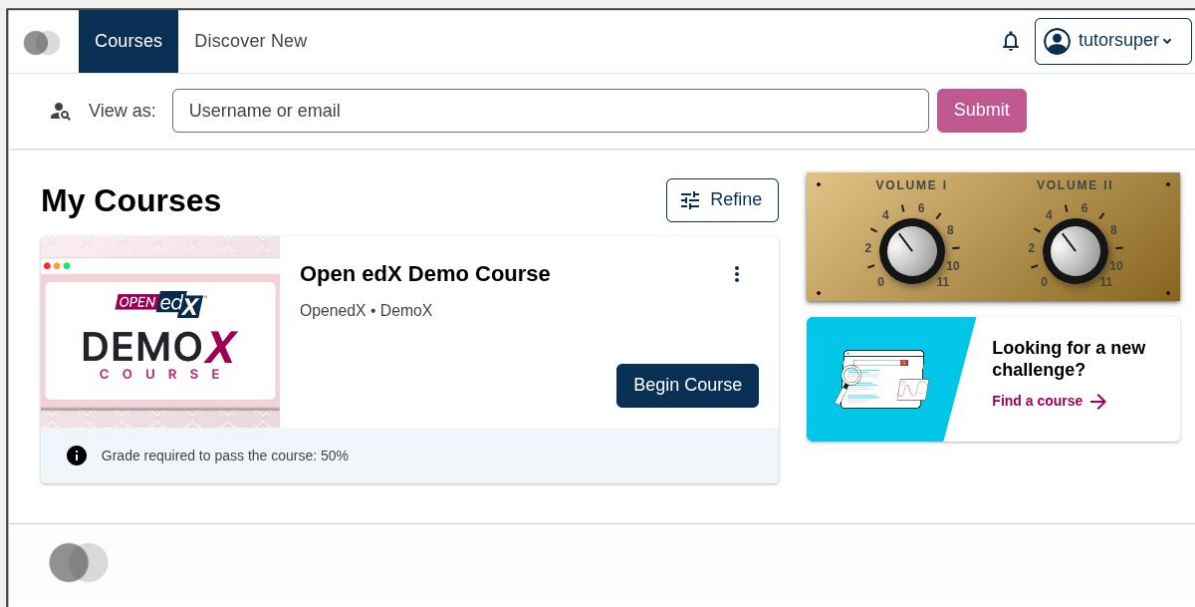
```
org.openedx.frontend.learner_dashboard.widget_sidebar.v1
```



<https://github.com/brian-smith-tcril/design-tokens-talk-2026/tree/main/plugin-learner-dashboard-sidebar-highlighter>

Application tokens in plugins

A plugin with knobs...



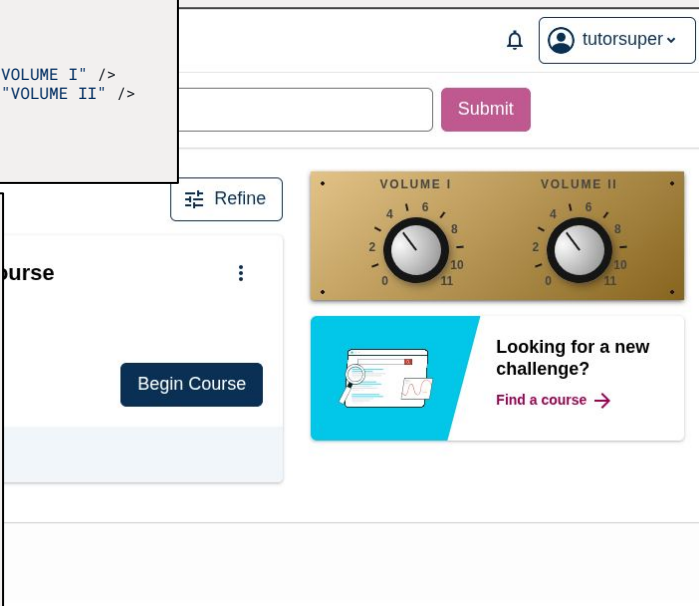
<https://github.com/brian-smith-tcril/design-tokens-talk-2026/tree/main/plugin-knobby>

Application tokens in plugins

A plugin with knobs...

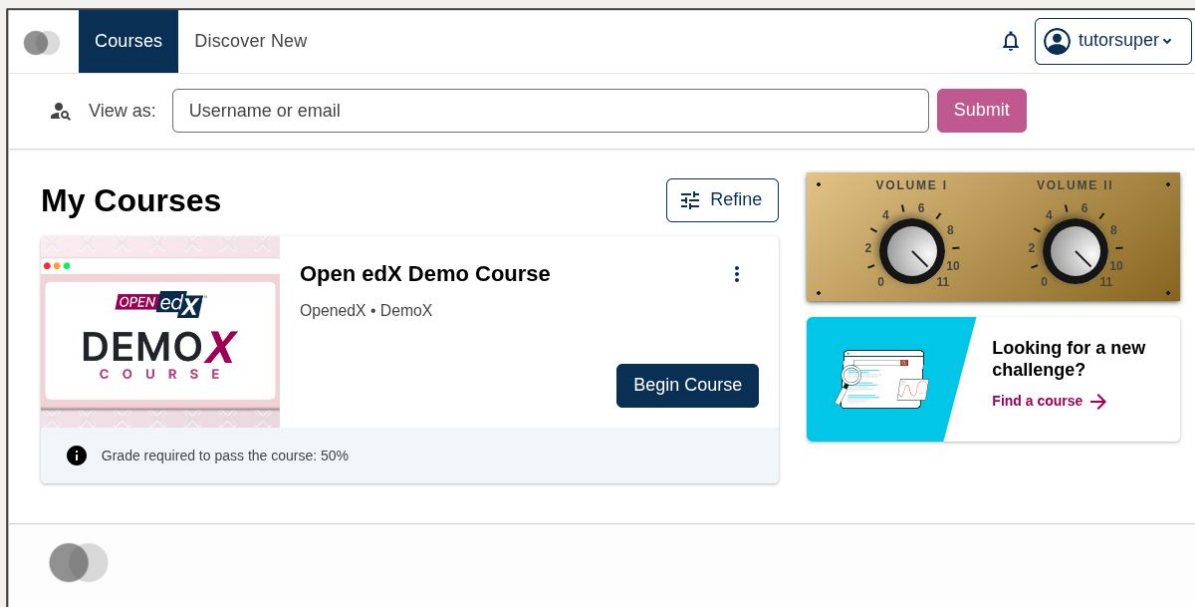
```
const Knobby = () => (  
  <...  
  <Knob value="--knobby-plugin-volume-i-knob, 4" label="VOLUME I" />  
  <Knob value="--knobby-plugin-volume-ii-knob, 4" label="VOLUME II" />  
  <...  
  </>  
);
```

```
const config = {  
  pluginSlots: {  
    'org.openedx.frontend.learner_dashboard.widget_sidebar.v1': {  
      keepDefault: true,  
      plugins: [  
        {  
          op: PLUGIN_OPERATIONS.Insert,  
          widget: {  
            id: 'knobby_plugin',  
            priority: 10,  
            type: DIRECT_PLUGIN,  
            RenderWidget: Knobby,  
          },  
        },  
      ],  
    },  
  },  
};  
  
export default config;
```



Application tokens in plugins

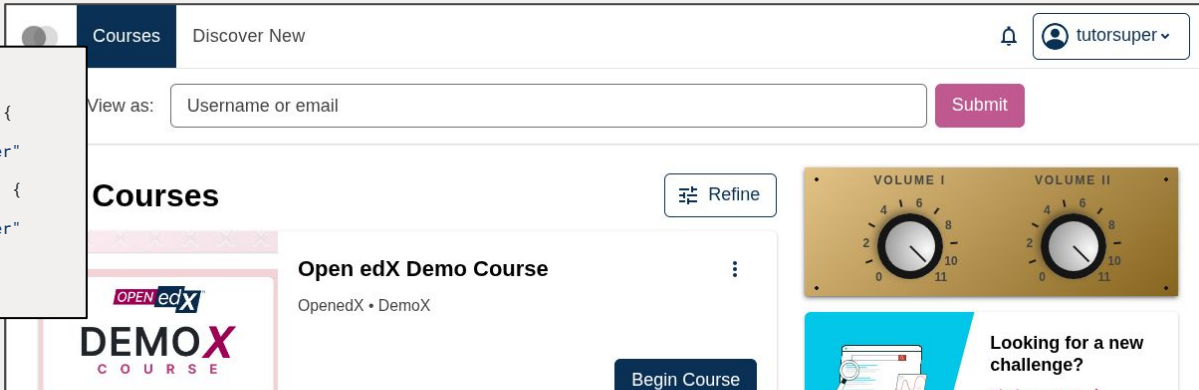
A plugin with knobs that brand packages can turn!



Application tokens in plugins

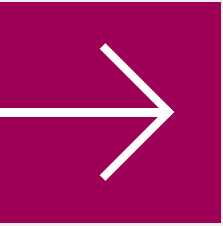
A plugin with knobs that brand packages can turn!

```
{
  "knobby-plugin": {
    "volume-i-knob": {
      "$value": "11",
      "$type": "number"
    },
    "volume-ii-knob": {
      "$value": "11",
      "$type": "number"
    }
  }
}
```



The screenshot shows the Open edX Demo Course interface. At the top, there's a navigation bar with 'Courses' and 'Discover New'. Below that, a search bar with 'View as:' and a 'Submit' button. The main content area features a 'Courses' section with a 'Refine' button. The 'Open edX Demo Course' is highlighted, showing a 'Begin Course' button. To the right, there are two volume knobs labeled 'VOLUME I' and 'VOLUME II', both set to 11. Below the knobs, there's a section titled 'Looking for a new challenge?' with a magnifying glass icon.

```
PARAGON_THEME_URLS: {
  variants: {
    light: {
      urls: {
        default: 'https://cdn.jsdelivr.net/npm/@openedx/paragon@latest/dist/light.min.css',
        brandOverride: 'https://cdn.jsdelivr.net/gh/brian-smith-tcril/design-tokens-talk-2026@[...]/brand-up-to-eleven/dist/light.min.css',
      },
    },
  },
},
```



Enabling customization

I found something I want to customize...

I found something I want to customize...

...and I can! There's a plugin slot for it!

I found something I want to customize...

...and I can! There's a plugin slot for it!

...and I can! There are tokens for it!

I found something I want to customize...

...and I can! There's a plugin slot for it!

...and I can! There are tokens for it!

...but sometimes I can't.

**I found something I want to customize...
...but I can't...**

I found something I want to customize...

...but I can't...

...because it's hardcoded



I found something I want to customize... ...but I can't...

...because it's hardcoded? **PR a bug fix!**

A photograph of a vintage cassette player. A pink arrow points from a text box on the left to the volume knob. The cassette player has a silver faceplate with three knobs labeled 'STEREO AMPLIFIER', 'BASS', and 'BALANCE'. Below the knobs are buttons for 'POWER', 'REC', 'PLAY', 'REW', 'FF', 'STOP/EJ', and 'PAUSE'. A cassette tape is partially inserted into the deck. The cassette tape has a label that says 'Metal'. The cassette player also has a 'COUNTER' display and a 'PHONES' jack.

```
--pgn-color-light-700
```

I found something I want to customize... ...but I can't...

...because it's hardcoded? **PR a bug fix!**

refactor: added new tokens



PKulkoRaccoonGang committed

on Aug 29, 2023

...

```
- margin-bottom: 1rem;  
+ margin-bottom: var(--pgn-spacing-spacer-base);
```

[https://github.com/openedx/paragon/pull/2568/changes/319f30\[...\]/diff-434df6\[...\]](https://github.com/openedx/paragon/pull/2568/changes/319f30[...]/diff-434df6[...])

I found something I want to customize...

...but I can't...

...because there's no knob



I found something I want to customize... ...but I can't...

...because there's no knob? **Add one in a PR!**



I found something I want to customize... ...but I can't...

...because there's no knob? **Add one in a PR!**

<https://github.com/openedx/frontend-app-authn/pull/69>

 Alert.jsx

```
- import Alert from '../logistration/Alert';  
+ import { Alert } from '@edx/paragon';
```


I found something I want to customize... ...but I can't...

...because there's no knob? **Add one in a PR!**

<https://github.com/openedx/frontend-app-authoring/pull/2837>

 **bradenmacdonald** approved these changes on Feb 5

[View reviewed changes](#)

bradenmacdonald left a comment Contributor ...

LGTM! Please just make the component colors more DRY by refactoring them to use CSS variables (not SCSS variables) and this is good to go as far as I'm concerned.

```
+  --content-library-component-primary-color: #005C9E;
```

```
[...]
```

```
.component-style-default {  
-  background-color: #005C9E;  
+  background-color: var(--content-library-component-primary-color);
```

I found something I want to customize...
...but I can't...

...because there's no plugin slot



Image via Amazon

I found something I want to customize...
...but I can't...

...because there's no plugin slot? **PR one!**



I found something I want to customize... ...but I can't...

...because there's no plugin slot? **PR one!**

feat: add plugin slot for content iframe error

component (#1771)

Verified

f91af21



jansenk authored on Oct 20, 2025 · 5 / 5

```
export const ContentIFrameErrorSlot : React.FC<Props> = ({ courseId }: Props) => (  
  <PluginSlot  
    id="org.openedx.frontend.learning.content_iframe_error.v1"  
    pluginProps={{ courseId }}  
  >  
    <ErrorPage />  
  </PluginSlot>  
)
```

```
- showError ? <ErrorPage /> : <ContentIFrameLoaderSlot courseId={courseId} loadingMessage={loadingMessage} />  
+ showError ? (  
+   <ContentIFrameErrorSlot courseId={courseId} />  
+ ) : (  
+   <ContentIFrameLoaderSlot courseId={courseId} loadingMessage={loadingMessage} />  
+ )
```

<https://github.com/openedx/frontend-app-learning/pull/1771/>



I found something I want to customize...
...but I can't...

...because my PR isn't landing!

I found something I want to customize... ...but I can't...

...because my PR isn't landing!



Too many knobs?

I found something I want to customize... ...but I can't...

...because my PR isn't landing!



Too many knobs?



Too many slots?

I found something I want to customize... ...but I can't...

...because my PR isn't landing!

Frontend extensibility discussion: documenting best practices

Development Architecture discussion frontend extensibility



brian.smith

3d

<https://discuss.openedx.org/t/frontend-extensibility-discussion-documenting-best-practices/19055>

I found something I want to customize... ...but I can't...

...because, as a community, we haven't decided on the best way to provide this extensibility

I found something I want to customize... ...but I can't...

...because, as a community, we haven't decided on the best way to provide this extensibility? **Fork and talk!**



Discussion Forum

Have Questions or Ideas? Let's Talk on the Forum

Join our discussion forum at discuss.openedx.org. We have categories for different audiences.

[Join the discussion](#)



Chat on Slack

Be Part of the Open edX Slack Community

Our real-time conversations are on Slack. You can request a Slack invitation, then join our community Slack workspace.

[Get Slack invitation](#)

**I found something I want to customize...
...but I can't...**

I found something I want to customize... ...but I can't...

Value hardcoded?

I found something I want to customize... ...but I can't...

Value hardcoded? Wire up a var.

I found something I want to customize... ...but I can't...

Value hardcoded? Wire up a var.

No token?

I found something I want to customize... ...but I can't...

Value hardcoded? Wire up a var.

No token? Add one.

I found something I want to customize... ...but I can't...

Value hardcoded? Wire up a var.

No token? Add one.

No plugin slot?

I found something I want to customize... ...but I can't...

Value hardcoded? Wire up a var.

No token? Add one.

No plugin slot? Add one.

I found something I want to customize... ...but I can't...

Value hardcoded? Wire up a var.

No token? Add one.

No plugin slot? Add one.

No community consensus?

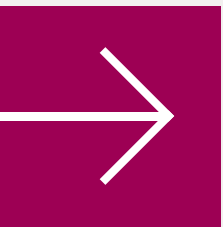
I found something I want to customize... ...but I can't...

Value hardcoded? Wire up a var.

No token? Add one.

No plugin slot? Add one.

No community consensus? Fork and talk.



Thank you!

Let's keep talking!

Slides and materials available

<https://github.com/brian-smith-tcrl/design-tokens-talk-2026>

